

Gaussian Prime Connectivity Transfer Operator (GPCTO): Methods for Scalable Moat Detection

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Abstract

The Gaussian moat problem asks whether, for each fixed step bound k , the connected component of the origin in the Gaussian prime proximity graph G_k is finite. We develop two complementary methods for investigating this question at radii far beyond current computational records. The **tile operator** compresses the connectivity of a rectangular region of the Gaussian lattice into boundary data—which connected components touch which faces—and admits a natural composition algebra. The **Independent Strip Ensemble (ISE)** deploys many collar-disjoint tile probes at the same radial distance; it is provably conservative (zero false negatives) with false-positive probability decaying exponentially in the number of strips. For rigorous proof, a **tile-based upper bound (UB)** method tiles the first-octant annulus gaplessly and checks global disconnection through composition. We further develop a **fat-stripe** method—a concrete realization of the UB approach—that tiles an annular strip in the first octant, processes each tile independently via row sieve and sparse union-find, composes all tiles via the GPCTO algebra, and delivers a definitive verdict on whether any connected component spans the annulus radially. Calibration against the known Tsuchimura moats at $k^2 = 26$ and $k^2 = 32$ confirms correct detection. At $k^2 = 36$, the ISE locates a percolation transition near $R \approx 80.4\text{M}$; at $k^2 = 40$, results through $R = 3.5\text{B}$ show a sigmoid collapse with $R_{0.5} \approx 839\text{M}$. Fat-stripe calibration confirms total annular barriers at $k^2 = 32$ ($R \approx 2.82\text{M}$) and $k^2 = 36$ ($R \approx 80\text{M}$), with cross-validation against ISE establishing consistency between the two methods. At $k^2 = 40$, five fat-stripe probes across $R \in [1.03\text{B}, 1.09\text{B}]$ detect total annular blockage—the first such detection at a k^2 value where no moat was previously known. Angular coverage is 128K lattice units ($\sim 0.007^\circ$); full-octant tiling is needed for rigorous proof.

1 Preliminaries

1.1 Gaussian integers

Definition 1.1 (Gaussian integers). The ring of Gaussian integers is $\mathbb{Z}[i] = \{a + bi : a, b \in \mathbb{Z}\}$, equipped with the squared norm $|z|^2 = a^2 + b^2$ for $z = a + bi$.

1.2 Gaussian primes

Definition 1.2 (Gaussian primes). An element $\pi \in \mathbb{Z}[i]$ is a *Gaussian prime* if it generates a prime ideal in $\mathbb{Z}[i]$. The classification is:

1. **Split primes.** If $p \equiv 1 \pmod{4}$ is a rational prime, then $p = a^2 + b^2$ for unique $a > b > 0$ (Cornacchia), and $\pi = a + bi$ and its associates are Gaussian primes with $|\pi|^2 = p$.
2. **Inert primes.** If $p \equiv 3 \pmod{4}$ is a rational prime, then p itself is a Gaussian prime with $|p|^2 = p^2$.

3. **Ramified prime.** The rational prime $2 = -i(1+i)^2$; the element $1+i$ is a Gaussian prime with $|1+i|^2 = 2$.

We write P for the set of all Gaussian primes. By the Gaussian prime number theorem, the count of Gaussian primes with norm at most N is asymptotically $C \cdot N / \ln N$, where C depends on convention. In a lattice rectangle at distance R from the origin, the local prime density is approximately

$$\rho(R) \approx \frac{C}{2 \ln R},$$

with $C \approx 1.274$ calibrated empirically.

1.3 The Gaussian prime graph

Definition 1.3 (Gaussian prime graph). Fix $k^2 \in \mathbb{N}$. The *Gaussian prime graph* is $G_k = (P, E_k)$ where

$$\{\pi, \pi'\} \in E_k \iff |\pi - \pi'|^2 \leq k^2.$$

The *seed set* is $\Sigma_k = \{\pi \in P : |\pi|^2 \leq k^2\}$, and the *origin component* $C_0(k)$ is the connected component of G_k containing every prime in Σ_k . (One may equivalently adjoin a formal vertex at the origin connected to all primes in Σ_k .)

1.4 The moat problem

Definition 1.4 (Moat radius). The *moat radius* is

$$R_{\text{moat}}(k) = \sup\{R \geq 0 : C_0(k) \text{ meets the circle } |z| = R\}.$$

The Gaussian moat problem asks: is $R_{\text{moat}}(k)$ finite for every fixed k ?

1.5 Prior art

k^2	R_{moat}	Source
2	$\sqrt{137} \approx 11.7$	Classical
6	$\approx 3,612$	Tsuchimura 2005
10	$\approx 26,862$	Tsuchimura 2005
18	$\approx 175,045$	Tsuchimura 2005
26	$\approx 1,015,639$	Tsuchimura 2005
32	$\approx 2,823,055$	Tsuchimura 2005
36	Unknown (ISE transition at $\approx 80\text{M}$)	This work
40	Unknown (ISE transition at $\approx 839\text{M}$)	This work

Table 1: Known and estimated moat radii.

Tsuchimura's method grows the origin component outward through all Gaussian primes, at cost $O(R^2 / \ln R)$ in both time and memory. This becomes prohibitive for $R > 10^8$.

2 The Tile Operator

2.1 Tile definition

Definition 2.1 (Tile). A *tile* is a lattice rectangle

$$T = T(a_0, b_0, W, H) = \{a + bi \in \mathbb{Z}[i] : a_0 \leq a \leq a_0 + H, b_0 \leq b \leq b_0 + W\},$$

where a is the radial coordinate and b the lateral coordinate. We call W the *width* and H the *height* of the tile.

2.2 The four faces

Definition 2.2 (Faces). The tile T has four distinguished faces. A Gaussian prime $\pi = a + bi \in P \cap T$ touches:

- the *Inner face* $\mathcal{I}$ if $a - a_0 \leq c$,
- the *Outer face* $\mathcal{O}$ if $(a_0 + H) - a \leq c$,
- the *Left face* $\mathcal{L}$ if $b - b_0 \leq c$,
- the *Right face* $\mathcal{R}$ if $(b_0 + W) - b \leq c$,

where c is the collar width defined below. These inequalities are non-strict ($\leq$, not $<$): a prime at distance exactly c from a face boundary is counted as touching that face.

2.3 Collar

Definition 2.3 (Collar). The *collar* is

$$c = \left\lceil \sqrt{k^2} \right\rceil.$$

The *expanded tile* is

$$T^+ = \{a + bi : a_0 - c \leq a \leq a_0 + H + c, b_0 - c \leq b \leq b_0 + W + c\}.$$

Justification. An integer step vector $(\Delta a, \Delta b)$ satisfying $\Delta a^2 + \Delta b^2 \leq k^2$ has maximal single-coordinate excursion

$$\max\{|\Delta a| : \exists \Delta b, \Delta a^2 + \Delta b^2 \leq k^2\} = \left\lfloor \sqrt{k^2} \right\rfloor.$$

The floor is the tight bound: the maximum $|\Delta a|$ is achieved at $\Delta b = 0$, giving $|\Delta a| \leq \lfloor \sqrt{k^2} \rfloor$.

Using $c = \lceil \sqrt{k^2} \rceil$ rather than $\lfloor \sqrt{k^2} \rfloor$ is the *conservative* choice. When k^2 is a perfect square, $\lceil \sqrt{k^2} \rceil = \lfloor \sqrt{k^2} \rfloor$, so the two definitions coincide. When k^2 is not a perfect square, $\lceil \sqrt{k^2} \rceil = \lfloor \sqrt{k^2} \rfloor + 1$, so the collar is one unit wider than strictly necessary.

This over-collaring is harmless for correctness—the expanded tile includes strictly more lattice points than required—and it provides two benefits:

1. **Uniform formula.** For any k^2 , the collar $c = \lceil \sqrt{k^2} \rceil$ guarantees that every edge incident to an interior prime is visible in T^+ , without case analysis on whether k^2 is a perfect square.
2. **Face-port completeness.** With $\leq$ in the face-membership inequalities, a prime at distance exactly c from the boundary is tagged as a face port. Using ceil ensures that this threshold is never below the actual maximum excursion, even though the excess is at most one unit.

Concrete values:

k^2	$\lfloor \sqrt{k^2} \rfloor$	$\lceil \sqrt{k^2} \rceil$	Remark
36	6	6	Perfect square; floor = ceil
40	6	7	Non-perfect square; collar oversized by 1

Table 2: Collar values for key step bounds.

Throughout this document, $c = \lceil \sqrt{k^2} \rceil$ is the canonical collar. All theorems are stated and proved using this definition.

2.4 Face ports

Definition 2.4 (Face ports). A *face port* is a record (π, C, F) where $\pi \in P \cap T$ is a Gaussian prime in the interior tile, C is the index of its connected component in $G_k[T^+]$, and $F \in \{\mathcal{I}, \mathcal{O}, \mathcal{L}, \mathcal{R}\}$ is the face it touches. A single prime may be a face port for multiple faces.

2.5 Tile connectivity function

Definition 2.5 (Tile connectivity function). Given tile T and step bound k^2 :

1. Form the induced subgraph $G_k[T^+]$ on all Gaussian primes in the expanded tile T^+ .
2. Compute the connected components of $G_k[T^+]$ via union-find.
3. For each component C , determine its *face set* $\partial_T C \subseteq \{\mathcal{I}, \mathcal{O}, \mathcal{L}, \mathcal{R}\}$ —the set of faces touched by primes of C that lie in the interior tile T .
4. The *transfer data* are the pairwise face-spanning counts:

$$\text{span}_T(X, Y) = \#\{C : X, Y \in \partial_T C\}, \quad X, Y \in \{\mathcal{I}, \mathcal{O}, \mathcal{L}, \mathcal{R}\}, \quad X \neq Y.$$

2.6 Inner-to-Outer count

Definition 2.6 (io_count). The *inner-to-outer count* is

$$\text{io_count}(T) = \text{span}_T(\mathcal{I}, \mathcal{O}),$$

the number of distinct connected components in $G_k[T^+]$ that touch both the Inner and Outer faces of T .

Interpretation. $\text{io_count}(T) > 0$ means there exists a path in $G_k[T^+]$ from a prime near the bottom edge to a prime near the top edge. $\text{io_count}(T) = 0$ means the tile is *radially blocked*: no connectivity passes through it in the outward direction.

3 Tile Composition

3.1 Horizontal composition

Definition 3.1 (Horizontal composition). Let $T_L = T(a_0, b_0, W, H)$ and $T_R = T(a_0, b_0 + s_b, W, H)$ be two tiles at the same radial level, separated laterally by stride s_b . Their *horizontal composition* $T_L \otimes_h T_R$ is a new tile operator whose components are obtained by:

1. Taking the union of all components from both tiles.
2. Merging component C_L from T_L with component C_R from T_R whenever there exist face ports $p_L \in \mathcal{R}(T_L)$ and $p_R \in \mathcal{L}(T_R)$ belonging to C_L and C_R respectively, with $|p_L - p_R|^2 \leq k^2$.
3. Recomputing face sets for the merged components.

The composed tile has faces $\mathcal{I}(T_L) \cup \mathcal{I}(T_R)$, $\mathcal{O}(T_L) \cup \mathcal{O}(T_R)$, $\mathcal{L}(T_L)$, and $\mathcal{R}(T_R)$.

3.2 Vertical composition

Definition 3.2 (Vertical composition). Let $T_B = T(a_0, b_0, W, H)$ and $T_T = T(a_0 + H, b_0, W, H)$ be two tiles stacked radially. Their *vertical composition* $T_B \otimes_v T_T$ merges components whenever there exist face ports $p_B \in \mathcal{O}(T_B)$ and $p_T \in \mathcal{I}(T_T)$ with $|p_B - p_T|^2 \leq k^2$. The composed tile has faces $\mathcal{I}(T_B)$, $\mathcal{O}(T_T)$, $\mathcal{L}(T_B) \cup \mathcal{L}(T_T)$, and $\mathcal{R}(T_B) \cup \mathcal{R}(T_T)$.

3.3 Composition soundness

Theorem 3.1 (Distance-based composition is conservative). *Let T_1, T_2 be tiles sharing a boundary, and let $T_1 \otimes T_2$ denote their composition via distance-based face-port matching. If two components C_1, C_2 are merged in $T_1 \otimes T_2$, then there exists a genuine edge $\{\pi_1, \pi_2\} \in E_k$ with $\pi_1 \in C_1, \pi_2 \in C_2$.*

Proof. The merge occurs because there exist face ports $p_1 \in C_1$ and $p_2 \in C_2$ with $|p_1 - p_2|^2 \leq k^2$. By Definition 1.3, this means $\{p_1, p_2\} \in E_k$. Since $p_1 \in C_1$ and $p_2 \in C_2$, the edge connects the two components in G_k . $\square$

Corollary 3.2. *Distance-based composition never introduces false connections: every merged pair of components is genuinely connected in G_k .*

Remark 3.3 (Conservative direction). Distance-based composition can, however, *miss* connections. Two primes $\pi_1 \in T_1$ and $\pi_2 \in T_2$ might be connected in G_k via a multi-hop path that passes through primes in the overlap region without either π_1 or π_2 being face ports. An alternative *shared-prime matching* scheme—identifying primes with identical coordinates in the overlap zone and merging their respective component assignments—captures all such connections exactly. The current implementation uses distance-based matching; shared-prime matching is formalized but not yet implemented. For the ISE method (Section 4), composition is not used and this distinction is immaterial. For the tile-UB method (Section 5), distance-based matching is sound (it can only undercount connections, which is conservative for proving disconnection) but potentially less tight than shared-prime matching.

4 Independent Strip Ensemble (ISE)

4.1 Strip definition

Definition 4.1 (Strip). A *strip* S_j at radial level R with tile dimensions $W \times H$ is a sequence of tiles:

$$S_j = \bigcup_{n=0}^{N-1} T(a_{l_0} + nH, b_j, W, H),$$

where a_{l_0} is chosen to place the inner face at radial distance R from the origin, b_j is the lateral offset of strip j , and $N = \lceil (R_{\max} - R_{\min})/H \rceil$ is the number of shells.

Definition 4.2 (Independence criterion). Two strips S_j and S_l are *independent* if their expanded tiles are disjoint in the lateral coordinate: the collar-expanded regions $[b_j - c, b_j + W + c]$ and $[b_l - c, b_l + W + c]$ do not overlap. This is guaranteed when the lateral stride satisfies

$$|b_j - b_l| \geq W + 2c.$$

The standard ISE layout places M strips at positive-only offsets:

$$b_j = c + j \cdot (W + 2c), \quad j = 0, 1, \dots, M - 1.$$

This ensures all M expanded tiles are laterally disjoint and avoids the $b \rightarrow -b$ reflection symmetry (Gaussian primality of $a + bi$ is identical to that of $a - bi$), which would halve the effective independent sample count.

4.2 The $f(r)$ metric

Definition 4.3 ($f(r)$). At radial shell n with midpoint $r_n = R_{\min} + (n + \frac{1}{2})H$, define

$$f(r_n) = \frac{\#\{j : \text{io_count}_j(r_n) > 0\}}{M},$$

where $\text{io_count}_j(r_n)$ is the inner-to-outer count of the tile in strip j at shell n .

Interpretation. $f(r)$ is the fraction of independent strip probes that detect Inner-to-Outer connectivity at radius r . It ranges from 0 (all strips blocked) to 1 (all strips crossing).

4.3 Subgraph monotonicity (zero false negatives)

Theorem 4.1 (Subgraph monotonicity). *Let $S \subseteq \mathbb{Z}[i]$ be any subset containing the seed set Σ_k . Let $R_{\text{strip}}(k, S)$ denote the moat radius of the origin component in the induced subgraph $G_k|_S$. Then*

$$R_{\text{strip}}(k, S) \leq R_{\text{moat}}(k).$$

Proof. The primes in S form a subset $P_S \subseteq P$. The graph $G_k|_S = (P_S, E_k|_{P_S})$ is an induced subgraph of G_k . Every path in $G_k|_S$ is also a path in G_k (each edge satisfies $|\pi - \pi'|^2 \leq k^2$ and both endpoints lie in P). Therefore every prime reachable from Σ_k in $G_k|_S$ is also reachable in G_k . Taking suprema of radii gives $R_{\text{strip}}(k, S) \leq R_{\text{moat}}(k)$. $\square$

Corollary 4.2 (Local crossing soundness). *If a tile T has $\text{io_count}(T) > 0$, then there exists a genuine Inner-to-Outer path in G_k through the collared region T^+ .*

Proof. The component witnessing $\text{io_count}(T) > 0$ lives in $G_k[T^+]$, which is a subgraph of G_k . The same path exists in G_k . $\square$

Theorem 4.3 (ISE zero false negatives). *Let $\{S_j\}_{j=0}^{M-1}$ be an ensemble of strips covering a radial interval $[R, R + H]$. If $f(r) = 0$ at every strip, then no path in G_k passes from $|z| < R$ to $|z| > R + H$ through any of the M sampled lateral positions.*

Proof. For each strip S_j , the tile at shell r computes $G_k[T_j^+]$, which contains every Gaussian prime in the expanded tile and every edge incident to any interior prime. If $\text{io_count}_j(r) = 0$, then no connected component of $G_k[T_j^+]$ spans from the Inner to the Outer face of T_j .

Any path in G_k from a prime with $a < a_{\text{lo}} + c$ to a prime with $a > a_{\text{lo}} + H - c$, passing through the lateral range $[b_j, b_j + W]$, must consist entirely of primes within T_j^+ (since each step moves at most c units laterally, and the strip width W exceeds $2c$, a path entering the strip's interior cannot exit the expanded tile's lateral range and re-enter within the same tile height). Therefore any such path would appear as a component in $G_k[T_j^+]$, contradicting $\text{io_count}_j = 0$. $\square$

Remark 4.4 (False positives are possible). $\text{io_count}(T) > 0$ certifies a local crossing but does not imply the crossing is part of the origin component $C_0(k)$. The ISE measures *local percolation* (whether any component crosses a shell), not *origin-connected reachability* (whether $C_0(k)$ crosses a shell). These are distinct: a Tsuchimura moat can exist at a radius where $f(r) > 0$, if the crossing components are disconnected from the origin.

4.4 False-positive analysis

Theorem 4.5 (Ensemble false-positive bound). *Suppose, at some shell r , the full graph G_k contains an Inner-to-Outer crossing. Let $p(r)$ denote the probability that a single strip fails to detect this crossing (i.e., $\text{io_count}_j(r) = 0$ despite the crossing existing in G_k). If the M strips are independent (collar-disjoint), then*

$$\Pr(f(r) = 0 \mid \text{crossing exists in } G_k) \leq p(r)^M.$$

Proof. The event $f(r) = 0$ requires all M strips to independently fail. Under the disjointness condition, the prime populations in distinct tiles are disjoint, so their `io_count` outcomes are determined by disjoint subsets of primes. The product bound follows. $\square$

Over N shells, the union bound gives $\Pr(\text{any false positive}) \leq N \cdot p(r)^M$. Concrete values:

p	$M = 16$	$M = 32$	$M = 64$
0.30	4.3×10^{-9}	1.9×10^{-17}	3.5×10^{-34}
0.10	10^{-16}	10^{-32}	10^{-64}
0.01	10^{-32}	10^{-64}	10^{-128}

Table 3: False-positive probability p^M for various single-strip failure probabilities.

Remark 4.6 (Independence model). The product formula requires that `io_count` outcomes across strips are independent. This holds exactly when the expanded tiles are laterally disjoint (guaranteed by $\text{stride} \geq W + 2c$). Gaussian primes are deterministic, so “independence” here means that the prime distribution in one tile is not predictive of the connectivity pattern in a distant tile. For tiles separated by $\gg \sqrt{k^2}$, this is empirically well-supported: the Hardy–Littlewood conjecture for Gaussian primes predicts asymptotic independence of prime counts in disjoint regions.

4.5 Curvature compensation

Problem. Strips are rectangular, but the moat is circular. Under naive placement (all strips sharing the same a_{lo}), a strip at lateral offset b_j has its Inner face center at Euclidean distance $\sqrt{a_{\text{lo}}^2 + (b_j + W/2)^2}$ from the origin. Lateral strips are therefore at slightly different radii.

Definition 4.4 (Compensated placement). For strip j with lateral center $b_{\text{center},j} = b_j + W/2$, set

$$a_{\text{lo},j} = \left\lfloor \sqrt{R_{\text{target}}^2 - b_{\text{center},j}^2} \right\rfloor,$$

where R_{target} is the desired inner-face radius for all strips.

Theorem 4.7 (Compensated placement accuracy). *Under compensated placement, the actual inner-face center distance from the origin satisfies*

$$\left| \sqrt{a_{\text{lo},j}^2 + b_{\text{center},j}^2} - R_{\text{target}} \right| < 1.$$

Proof. By definition, $a_{\text{lo},j} = \lfloor x \rfloor$ where $x = \sqrt{R_{\text{target}}^2 - b_{\text{center},j}^2}$. Then $x - 1 < a_{\text{lo},j} \leq x$, so

$$a_{\text{lo},j}^2 + b_{\text{center},j}^2 \leq x^2 + b_{\text{center},j}^2 = R_{\text{target}}^2,$$

and

$$a_{\text{lo},j}^2 + b_{\text{center},j}^2 > (x - 1)^2 + b_{\text{center},j}^2 = R_{\text{target}}^2 - 2x + 1.$$

Since $x > 0$, the squared distance lies in $(R_{\text{target}}^2 - 2x + 1, R_{\text{target}}^2]$, giving a radial error bounded by $1/R_{\text{target}} < 1$ for $R_{\text{target}} > 1$. $\square$

Remark 4.8. Compensated placement preserves strip independence: only the a -coordinate changes per stripe; the lateral separation $|b_j - b_l| \geq W + 2c$ is unchanged.

4.6 Angular coverage and stripe count

The ISE covers an angular sector of the first quadrant from $\theta = 0$ (the positive real axis) to $\theta_{\max} = \arctan(b_{\max}/R)$, where $b_{\max} = c + (M-1)(W+2c) + W$ is the rightmost lateral extent.

For $M = 32$, $W = 2000$, $c = 7$ (at $k^2 = 40$): $b_{\max} = 7 + 31 \cdot 2014 + 2000 = 64,441$. At $R = 1B$: $\theta_{\max} \approx 0.0037^\circ$. The angular coverage is a tiny fraction of the full annulus, which is precisely the point: the ISE does not need angular coverage. It needs M independent samples, and by the p^M bound, even $M = 32$ collar-disjoint samples suffice for exponentially reliable detection.

5 Tile-Based Upper Bound (UB) Method

5.1 Concept

The ISE detects moat *candidates* but cannot rigorously prove a moat exists: it samples only a sparse set of angular positions and measures local percolation rather than origin-connected reachability. To *prove* $R_{\text{moat}}(k) \leq R$ for some R , one must demonstrate that no path in G_k crosses from $|z| < R$ to $|z| > R + H$ through any angular position.

5.2 Octant symmetry

The dihedral symmetry group D_4 of $\mathbb{Z}[i]$ —generated by the quarter-turn $z \mapsto iz$ and conjugation $z \mapsto \bar{z}$ —preserves Gaussian primality. Consequently, it suffices to verify disconnection in the *first octant*: the region $\{a + bi : 0 \leq b \leq a\}$.

Lemma 5.1 (Octant barrier implies full-plane moat). *If no connected component of G_k restricted to the first-octant rectangular band $\{a + bi : R_{\min} \leq a \leq R_{\max}, 0 \leq b \leq a\}$ has a component crossing from $a = R_{\min}$ to $a = R_{\max}$, then $R_{\text{moat}}(k) \leq R_{\max}\sqrt{2}$.*

Proof sketch. Any path from the origin to $|z| > R_{\max}\sqrt{2}$ must eventually have $\max(|a|, |b|) > R_{\max}$, hence $a > R_{\max}$ in at least one of the eight octants (by the octant symmetries). The rectangular barrier in the first octant, transported to all eight octants by D_4 , blocks any such path. $\square$

Remark 5.2. The first-octant restriction is conservative for the UB: connections crossing the octant boundaries ($b = 0$ and $b = a$) are ignored, which can only undercount connectivity.

5.3 Gapless tiling geometry

Definition 5.1 (Gapless tiling). A *gapless tiling* of the first-octant annulus at radius R is a collection of tiles $\{T_{n,j}\}$ such that:

1. Every Gaussian prime π with $R_{\min} \leq \text{Re}(\pi) \leq R_{\max}$ and $0 \leq \text{Im}(\pi) \leq \text{Re}(\pi)$ lies in at least one tile's interior.
2. Tiles in the same radial shell n are spaced with lateral stride $s_b = W$ (not $W + 2c$), so tile interiors partition the band laterally.
3. Adjacent tiles share expanded regions of width $2c$ in the overlap zone.

Definition 5.2 (Tiles per radial shell). At radius R , the first-octant lateral extent is approximately R (from $b = 0$ to $b = a \approx R$). The number of tiles per shell is

$$N_\ell = \left\lceil \frac{R}{W} \right\rceil.$$

At $R = 1.5B$, $W = 2000$: $N_\ell = 750,000$.

5.4 Composition for UB

Tiles in each radial shell are composed horizontally (Left/Right matching) to produce a *band operator* for the entire first-octant width. Then consecutive radial shells are composed vertically (Inner/Outer matching). If the final composed operator has `io_count = 0` for the full octant width, the moat is confirmed at radius R .

The composition uses distance-based face-port matching (Theorem 3.1), which is sound for the UB direction: it can only undercount connections, making “blocked” verdicts reliable.

5.5 ISE-to-UB handoff

The ISE provides two quantities that guide the UB campaign:

1. $R_{0.5}$: the radius where $f(r) = 0.5$ (the percolation transition midpoint).
2. R_{ext} : the radius where $f(r) \approx 0$ (the extinction radius).

The UB campaign starts at R_{ext} , where ISE predicts disconnection is most likely. This avoids a blind search from the origin: instead of scanning all radii, start where the signal indicates the moat is.

5.6 Computational feasibility

5.6.1 Current `tile_cuda` architecture

The `tile_cuda` binary splits tile processing across GPU and CPU. Measured on a Jetson Orin Nano with 2000×2000 tiles at $k^2 = 36$:

Stage	Device	Time
GPU primality (sieve + Miller–Rabin)	GPU	~ 170 ms
Device-to-host bitmap copy	DMA	~ 2 ms
CPU union-find + face classification	CPU	~ 40 ms
CUDA context init (one-time)	GPU	~ 150 ms
Total per tile (after init)		~ 330 ms

Table 4: Per-tile timing breakdown on Jetson Orin Nano (`tile_cuda`).

5.6.2 GPU union-find gap

The current implementation runs union-find on CPU. The `gpu_cc_ms` field exists in the JSON output but is always 0.0—GPU union-find was planned but not implemented. The header `tile_kernel.cuh` is named suggestively but contains a primality kernel from an earlier iteration, not a union-find kernel.

Moving union-find to GPU (using ECL-CC [1] or Afforest [2] algorithms) eliminates the device-to-host bitmap copy and the CPU serialization bottleneck. The tile grid (2000×2000 , ~ 4 M cells) fits comfortably in GPU shared memory or L2 cache, and the connectivity pattern (128 neighbors for $k^2 = 40$) produces a sparse but regular graph.

5.6.3 Tile-per-SM batching

A single 2000×2000 tile generates $\sim 2,013$ CUDA blocks. On a 4090 with 128 SMs, this underutilizes the GPU (only ~ 16 blocks/SM). With a fully GPU-side pipeline, multiple tiles can be processed concurrently—one tile per SM—achieving full hardware utilization.

5.6.4 Throughput projections

Hardware	SMs	Clock (MHz)	Per-tile (batched)	589K tiles
Jetson Orin Nano	16	918	~ 180 ms	~ 29 hr
RTX 4090	128	2520	~ 1.8 ms	~ 18 min
A100 SXM4	108	1410	~ 2.2 ms	~ 22 min
H100 SXM5	132	1980	~ 1.5 ms	~ 15 min

Table 5: Projected throughput with fully-GPU tile pipeline and tile-per-SM batching. The 589K tiles column shows the wall time for a single-shell UB at $R = 1.5B$ (first-octant coverage).

5.6.5 UB campaign cost

Scenario	Tiles	1×4090	8×4090	$8 \times A100$
Single shell ($R = 1.5B$)	589K	18 min	2.3 min	2.8 min
1M radial (worst case)	294.5M	8.5 days / \$408	25 hr / \$400	11 hr / \$176

Table 6: UB campaign cost projections with GPU union-find and tile-per-SM batching.

5.6.6 CPU bottleneck on large GPUs

On fast GPUs, the CPU union-find becomes the bottleneck even with batching: at $40 \text{ ms/tile} \times 589K \text{ tiles}/n_{\text{cores}}$, a 64-core CPU still serializes to ~ 6.1 min for the UF step alone—comparable to the GPU primality time on a 4090. A fully GPU-side pipeline eliminates this entirely; no bitmap leaves the device until the final face-port summary.

5.6.7 Implementation status

GPU union-find for grid graphs is a bounded engineering task using well-studied algorithms (ECL-CC, Afforest). The gap is documented; the method’s computational feasibility does not depend on novel GPU research. The `gpu_cc_ms` field provides a natural integration point for timing instrumentation once the kernel is implemented.

5.7 Status

The UB method is formalized but not yet deployed as a campaign. The existing `compose.rs` implements distance-based matching (sound for UB). Shared-prime matching, which would yield tighter composition, is specified but not yet implemented. Both adversarial reviews (Sections on tile-UB and ISE challenges) returned verdicts of “Accept with revisions.”

6 Fat Stripe: Tile-Composed Annular Moat Detection

The fat-stripe method is a concrete realization of the tile-based UB approach (Section 5). Rather than leaving gapless tiling as a feasibility sketch, fat stripe implements it end-to-end: tile the first-octant annular strip, process each tile independently, compose all tiles via the GPCTO algebra, and read off a definitive verdict on whether any connected component of G_k spans the annulus radially. Unlike the ISE, which samples sparse angular positions and measures local percolation, fat stripe covers the full first-octant angular extent and answers a strictly stronger question: *does any component—origin-connected or not—bridge the annular band?*

6.1 Annular strip geometry

Definition 6.1 (Annular strip). Fix radii $r_{\min} < r_{\max}$ and step bound k^2 . The *first-octant annular strip* is the lattice region

$$\mathcal{A} = \{a + bi \in \mathbb{Z}[i] : r_{\min} \leq a \leq r_{\max}, 0 \leq b \leq b_{\max}(a)\},$$

where $b_{\max}(a) = \min(\lceil \sqrt{r_{\max}^2 - a^2} \rceil, a)$ enforces both the circular outer boundary and the first-octant constraint $b \leq a$.

Remark 6.1 (Angular span). The angular half-width of $\mathcal{A}$ at radius R is approximately

$$\theta \approx \arctan\left(\frac{\sqrt{2R \cdot \Delta r}}{R}\right) \approx \sqrt{\frac{2\Delta r}{R}},$$

where $\Delta r = r_{\max} - r_{\min}$. For $\Delta r/R \sim 10^{-4}$ (a 10K-wide annulus at $R = 80\text{M}$), $\theta \approx 0.8^\circ$, covering roughly 11% of the first octant. The 8-fold exact symmetry of $\mathbb{Z}[i]$ under the dihedral group D_4 (Lemma 5.1) ensures that a barrier in this sector implies a barrier in all eight octants, provided the full octant angular range is covered; see Section 6.7 for a discussion of partial-octant coverage.

6.2 Tiling and tile processing

The annular strip $\mathcal{A}$ is partitioned into a grid of tiles. The radial dimension is divided into *stripes* of height H (typically $H = W$, the tile width), and each stripe is divided laterally into tiles of width W .

Definition 6.2 (Radial stripe). A *radial stripe* is the sub-band

$$\mathcal{A}_n = \{a + bi \in \mathcal{A} : a_n \leq a < a_n + H\},$$

where $a_n = r_{\min} + nH$ for $n = 0, 1, \dots, N_r - 1$ and $N_r = \lceil (r_{\max} - r_{\min})/H \rceil$.

Within each stripe, tiles are indexed laterally: $T_{n,j} = T(a_n, b_j, W, H)$ with $b_j = jW$, for $j = 0, 1, \dots, J_n - 1$ where $J_n = \lceil b_{\max}(a_n)/W \rceil$.

Each tile is processed independently in three phases:

1. **Row sieve + Miller–Rabin.** For each row (fixed a) of the expanded tile $T_{n,j}^+$, a sieve with splitting primes up to $L = 110,000$ eliminates composite Gaussian norms, and deterministic Miller–Rabin (4–12 witnesses depending on norm magnitude) confirms surviving candidates. This produces a prime bitmap for the expanded tile.
2. **Sparse union-find.** Primes are compacted from the bitmap into a list of approximately $P \approx 4S^2/(\pi \ln n)$ entries (where $S = W + 2c$ is the expanded tile side and n is the typical norm). A rank table constructed via block-wise popcount enables $O(1)$ neighbor-index lookup. Union-find over the compacted list—checking all backward offset vectors $(\Delta a, \Delta b)$ with $\Delta a^2 + \Delta b^2 \leq k^2$ —identifies connected components. Working set: approximately 500 KB (rank table + UF arrays), versus 16 MB for a dense UF over all lattice points.
3. **Face-port extraction.** For each connected component, the algorithm records which faces ($\mathcal{I}, \mathcal{O}, \mathcal{L}, \mathcal{R}$) it touches within the interior (non-collar) tile, together with the (a, b) coordinates of the face-port primes and their component identifiers. This produces a `TileOperator` as defined in Section 2.

6.3 Hierarchical composition

Tile operators are composed in a two-level hierarchy that mirrors the strip-and-tile geometry:

Step 1: Horizontal composition within stripes. Within each radial stripe $\mathcal{A}_n$, tiles are composed left-to-right via $\otimes_h$ (Definition 3.1). For efficiency, tiles are grouped into *column-chunks* of C consecutive tiles (typically $C = 200$); tiles within each chunk are composed sequentially, then inter-chunk seams are merged. The result is a single band operator B_n for each radial stripe.

Step 2: Vertical composition across stripes. Band operators are composed bottom-to-top via $\otimes_v$ (Definition 3.2), producing a single operator B for the full annular strip.

Theorem 6.2 (Composition order independence). *The final composed operator B is independent of the order in which horizontal and vertical compositions are evaluated.*

Proof. By Theorem 3.1, each composition step merges components via union-find on face-port distances. Union-find is commutative and associative: the final partition depends only on the set of union operations, not their order. Any evaluation schedule—left-to-right within rows then bottom-to-top across rows, binary reduction trees, or any hybrid—produces the same set of face-port pairs satisfying $|p_1 - p_2|^2 \leq k^2$, hence the same final partition. $\square$

6.4 Spanning verdict

Definition 6.3 (Radial spanning). A connected component C of the composed operator B is *radially spanning* if there exist face ports $p_{\text{in}}, p_{\text{out}} \in C$ such that

$$|p_{\text{in}}| \leq r_{\min} + c \quad \text{and} \quad |p_{\text{out}}| \geq r_{\max} - c,$$

where $|\cdot|$ denotes the Euclidean norm $\sqrt{a^2 + b^2}$ and c is the collar width.

Remark 6.3 (Radius-based vs. face-based verdict). Because the tiling is Cartesian while the moat is circular, the Inner and Outer face labels of the composed operator do not correspond exactly to the circular boundaries $|z| = r_{\min}$ and $|z| = r_{\max}$. The spanning verdict therefore inspects the actual (a, b) coordinates of all face ports, computing $\sqrt{a^2 + b^2}$ for each, rather than relying on the rectangular face assignments. This avoids false verdicts near the corners of the tiled region where the circular and rectangular boundaries diverge.

Theorem 6.4 (Soundness of the spanning verdict). *If no component of B is radially spanning (Definition 6.3), then no connected component of G_k restricted to the first-octant annular strip $\mathcal{A}$ has a path from $|z| \leq r_{\min}$ to $|z| \geq r_{\max}$.*

Proof. Suppose for contradiction that a path $\pi_1, \pi_2, \dots, \pi_m$ in $G_k[\mathcal{A}]$ connects some π_1 with $|\pi_1| \leq r_{\min}$ to π_m with $|\pi_m| \geq r_{\max}$. Every edge $\{\pi_i, \pi_{i+1}\}$ satisfies $|\pi_i - \pi_{i+1}|^2 \leq k^2$ and both endpoints lie in $\mathcal{A}$.

By the gapless tiling (Definition 5.1), every prime in $\mathcal{A}$ belongs to the interior of at least one tile $T_{n,j}$. Each consecutive pair (π_i, π_{i+1}) either lies in the same expanded tile (captured by the local UF) or straddles a tile boundary (captured by composition, since both endpoints are face ports in their respective tiles). Therefore, all π_i lie in the same component of B .

The first prime π_1 has $|\pi_1| \leq r_{\min}$, making it a face port satisfying $|p| \leq r_{\min} + c$ (it lies in the face region of the lowest stripe). The last prime π_m has $|\pi_m| \geq r_{\max}$, making it a face port satisfying $|p| \geq r_{\max} - c$. Hence their shared component is radially spanning—contradicting the assumption. $\square$

Corollary 6.5 (Conservative direction). *The fat-stripe verdict “blocked” (no spanning component) is conservative: it can only be declared when no component of G_k crosses the annulus within the tiled region. False “blocked” verdicts are impossible; false “not blocked” verdicts can occur only due to missed connections from distance-based composition (Remark 3.3).*

6.5 Moat types and the percolation regime

The ISE (Section 4) and fat stripe answer related but distinct questions. The ISE measures whether *any* component crosses a single tile’s radial extent; fat stripe measures whether *any* component spans an entire annular strip. These yield different detection capabilities depending on the percolation regime of G_k at the candidate moat radius.

Definition 6.4 (Total barrier). A moat at radius R is a *total barrier* if no connected component of G_k —whether origin-connected or not—has a path from $|z| < R$ to $|z| > R + \Delta r$, for some annular width $\Delta r > 0$.

Definition 6.5 (Origin-only moat). A moat at radius R is *origin-only* if the origin component $C_0(k)$ does not reach $|z| > R$, but other connected components of G_k do have paths crossing the annular region around R .

Whether a moat is a total barrier or origin-only depends on the local percolation parameters:

k^2	R_{moat}	Bkwd offsets	Density ρ	$E[\text{bkwd deg}]$	$E[\text{total deg}]$	Moat type
2	~ 12	2	0.225	0.45	0.90	Total barrier
4	~ 45	2	0.153	0.31	0.61	Total barrier
26	$\sim 1.02\text{M}$	22	0.045	0.99	1.98	Origin-only
32	$\sim 2.82\text{M}$	26	0.042	1.09	2.18	Total barrier
36	$\sim 80\text{M}$	26	0.034	0.89	1.79	Total barrier
40	$\sim 1.05\text{B}$	30	0.030	0.90	1.81	Expected total

Table 7: Percolation parameters and moat type classification at known and estimated moat radii.

At $k^2 = 26$, the expected total degree at the moat radius is approximately 2.0, placing the graph in the supercritical regime of site percolation on a random geometric graph. In this regime, large-scale connected components freely bridge any finite-width annular band. Empirically, a 30-unit-wide band at the Tsuchimura moat for $k^2 = 26$ contains 41 independent spanning components. Fat stripe reports “not blocked” for such moats, correctly reflecting that the annulus is not a total barrier.

At $k^2 = 32$ and $k^2 = 36$, the total degree drops to 1.8–2.2 at the moat radius. The graph is sufficiently sparse that large-scale fragmentation prevents *any* component from spanning the annulus, not merely the origin component. Fat stripe detects these as total barriers.

Remark 6.6 (Complementarity with ISE). For origin-only moats ($k^2 = 26$), the ISE detects $f(r)$ depressions but not $f(r) = 0$, because orphan clusters cross the annulus. Fat stripe confirms “not blocked.” For total-barrier moats ($k^2 = 32, 36$), both methods agree: ISE shows the percolation transition, and fat stripe confirms absolute disconnection. The two methods are complementary: ISE scans efficiently for moat candidates across wide radial ranges; fat stripe provides the definitive verdict at specific candidate radii.

6.6 Calibration results

Fat-stripe calibration runs were performed on a Mac Mini (M-series) using the **fat-stripe** crate with $L = 110,000$ sieve limit, deterministic Miller–Rabin, and sparse union-find.

6.6.1 $k^2 = 32$ at $R \approx 2.82\text{M}$

The known Tsuchimura lower bound for $k^2 = 32$ is $R_{\text{moat}} \approx 2,823,054$ (the origin component reaches this radius but no further).

Parameter	Value
Annulus	[2,820,000, 2,830,000] (width 10,000)
Tile size	2000×2000
Radial stripes	5
Total tiles	449
Column-chunk size	200
Angular span	4.82° ($\approx 11\%$ of first octant)
Elapsed time	34 seconds
Verdict	BLOCKED — 0 spanning components

Table 8: Fat-stripe calibration at $k^2 = 32$, $R \approx 2.82\text{M}$.

6.6.2 $k^2 = 36$ at $R \approx 80\text{M}$

The known Tsuchimura upper bound for $k^2 = 36$ is $R_{\text{moat}} < 80,015,782$.

Parameter	Value
Annulus	[80,005,000, 80,015,000] (width 10,000)
Tile size	2000×2000
Radial stripes	5
Total tiles	2,373
Column-chunk size	200
Elapsed time	4 minutes 5 seconds
Verdict	BLOCKED — 0 spanning components

Table 9: Fat-stripe calibration at $k^2 = 36$, $R \approx 80\text{M}$.

Both calibration runs confirm that at the known moat radii for $k^2 = 32$ and $k^2 = 36$, the annular strip is a total barrier: no connected component of G_k spans from the inner to the outer radius within the tiled first-octant region.

6.6.3 $k^2 = 40$ at $R \in [1.03\text{B}, 1.09\text{B}]$

No moat at $k^2 = 40$ was previously known. The ISE sigmoid (Section 7.3) with $R_{0.5} \approx 839\text{M}$ and the percolation parameter analysis (Section 6.8) predicted that the candidate moat radius lies near $R \approx 1.05\text{B}$ in the same fragmentation regime as $k^2 = 36$. Five fat-stripe probes were run across a 60M radial band to test this prediction.

Each probe covers a $128,000 \times 128,000$ annular strip (64×64 tiles at $W = 2000$), with angular width $b_{\text{max}} = 128,000$.

Probe	R	Spanning components	Tiles	Time (Mac)
1	1.03B	0	4,096	5m 15s
2	1.04B	0	4,096	5m 15s
3	1.05B	0	4,096	5m 13s
4	1.07B	0	4,096	5m 12s
5	1.09B	0	4,096	5m 9s
Verdict	BLOCKED — 0 spanning components (all probes)			

Table 10: Fat-stripe probes at $k^2 = 40$, $R \in [1.03\text{B}, 1.09\text{B}]$.

Per-tile throughput: ~ 0.076 s/tile on Mac (M-series), comparable to the $k^2 = 36$ calibration run.

Angular coverage. At $R = 1B$, the 128K angular width subtends $\sim 0.007^\circ$, or $\sim 0.016\%$ of the first octant. This is a narrow-wedge probe—sufficient to detect the fragmentation regime and confirm methodological consistency with $k^2 = 32$ and $k^2 = 36$, but far below the full-octant coverage (Lemma 6.7) required for a rigorous moat proof.

Significance. This is the first detection of total annular blockage at $k^2 = 40$. Unlike $k^2 = 32$ and $k^2 = 36$, where the moat was independently known from Tsuchimura’s exhaustive method, $k^2 = 40$ has no prior moat result. The fat-stripe probes advance from calibration (confirming known results) to discovery (detecting a new candidate moat region).

6.6.4 Cross-validation with ISE at $k^2 = 32$

ISE probes with 200×200 tiles and 32 strips at the same radial location show per-tile $f(r)$ values ranging from 0.31 to 0.78—individual small tiles have 31–78% inner-to-outer connectivity. This is *not* contradictory with the fat-stripe “blocked” verdict. Individual tiles are well-connected *locally* (most components within a single tile span its 200-unit radial extent), but *long-range* radial connectivity through the full 10,000-unit composed strip breaks down at the moat distance. The moat is a large-scale fragmentation phenomenon, not a local one: components that bridge a single tile’s height become disconnected from one another when the full angular extent is considered.

6.7 Geometry and symmetry

Partial-octant coverage. The angular span of the fat-stripe annulus at $R = r$ with width Δr is approximately $\theta \approx \sqrt{2\Delta r/R}$, which for the calibration runs amounts to 5–11% of the first octant. The full octant spans 45° , so the calibrated region covers roughly $1\text{--}5^\circ$ out of 45° .

Lemma 6.7 (Octant symmetry for total barriers). *Let $\mathcal{A}_\theta$ denote the first-octant annular strip restricted to angular positions $[0, \theta]$. If no component of G_k restricted to $\mathcal{A}_\theta$ spans from $r_{\min}$ to $r_{\max}$, this does not immediately imply a full-annulus barrier. However, by the 8-fold dihedral symmetry of $\mathbb{Z}[i]$, an identical argument applies to each of the 8 octant copies of $\mathcal{A}_\theta$. A total annular barrier is proven if and only if: (a) the tiled region covers the full first-octant angular extent ($0 \leq b \leq a$), or (b) separate fat-stripe runs cover complementary angular sectors whose union tiles the full octant gaplessly.*

Remark 6.8 (Calibration vs. proof). The calibration results of Section 6.6 demonstrate that fat stripe correctly identifies total barriers at known moat radii ($k^2 = 32, 36$) and detects a new candidate barrier ($k^2 = 40$), but all probes tile only a sub-sector of the first octant. A rigorous moat proof requires full-octant coverage (Lemma 6.7). For $k^2 = 32$ and $k^2 = 36$, the sub-sector verdicts confirm methodological correctness against independently verified Tsuchimura moats. For $k^2 = 40$, the narrow-wedge probes ($\sim 0.007^\circ$) demonstrate that total blockage occurs in the predicted fragmentation regime, but full-octant tiling is required for a rigorous moat claim.

6.8 Implications for $k^2 = 40$

The percolation parameters at the expected moat radius for $k^2 = 40$ ($R \approx 1.05B$) closely resemble those at the confirmed $k^2 = 36$ moat:

The near-identical total degree (1.79 vs. 1.81) and similar backward degree suggest that $k^2 = 40$ at its moat radius occupies the same percolation regime as $k^2 = 36$: sufficiently sparse for large-scale fragmentation to create a total barrier. This prediction is now confirmed by the

Parameter	$k^2 = 36$ at $R = 80\text{M}$	$k^2 = 40$ at $R = 1.05\text{B}$
Local prime density ρ	0.034	0.030
$E[\text{backward degree}]$	0.89	0.90
$E[\text{total degree}]$	1.79	1.81

Table 11: Percolation parameter comparison between $k^2 = 36$ and $k^2 = 40$ at their respective moat radii.

fat-stripe probes of Section 6.6.3: five probes across $R \in [1.03\text{B}, 1.09\text{B}]$ show zero spanning components, consistent with the ISE sigmoid ($R_{0.5} \approx 839\text{M}$) predicting complete fragmentation by $R \approx 1\text{B}$.

Scaling considerations. A full-octant fat-stripe campaign at $R \approx 1.05\text{B}$ with $\Delta r = 10,000$ requires approximately $J = \lceil 1.05 \times 10^9 / 2000 \rceil = 525,000$ tiles per stripe and $N_r = 5$ radial stripes, totaling $\sim 2.6\text{M}$ tiles. At the calibrated throughput of ~ 0.5 s/tile (Mac Mini), this amounts to approximately 360 hours serial. With Rayon parallelism across 8–64 cores and a double-buffered pipeline, wall-clock time reduces to 6–45 hours depending on hardware. Cloud deployment on multi-core instances makes a single-day campaign feasible.

7 Empirical Validation

7.1 Calibration against known moats

ISE calibration runs were performed on a Jetson Orin Nano with $M = 32$ positive-only strips, square tiles, and the scanline kernel (Montgomery multiplication, row-sieve, union-find).

k^2	Tile size	Radial range	Shells	Known moat R	ISE $f(r)$ at moat	Control $\bar{f}$	Moat sweep $\bar{f}$
2	8×8	$[0, 100]$	13	~ 12	0.000 at $R=20$	—	0.19
26	500×500	$[950\text{K}, 1.1\text{M}]$	300	1,015,639	0.250 at $R \approx 1,016,250$	0.937	0.477
32	500×500	$[2.7\text{M}, 2.9\text{M}]$	400	2,823,055	0.344 at $R \approx 2,823,250$	0.891	0.503
36	2000×2000	$[79.5\text{M}, 80.5\text{M}]$	500	Unknown	0.000 at $R=80,399,000$	0.881	0.212

Table 12: ISE calibration against known Tsuchimura moats.

Key findings:

1. $k^2 = 2$: ISE correctly detects the moat with six $f(r) = 0$ shells spanning $R = 20$ –76.
2. $k^2 = 26, 32$: ISE detects $f(r)$ depressions near the Tsuchimura moats but does not reach $f(r) = 0$. This is expected: the moat is a structural gap in the origin component, not a collapse of all local connectivity. Orphan clusters provide local IO crossings even at the moat radius. The ISE correctly distinguishes control regions ($\bar{f} > 0.88$) from moat regions ($\bar{f} \approx 0.50$).
3. $k^2 = 36$: A single $f(r) = 0$ shell at $R = 80,399,000$ with monotonic approach ($f = 0.156 \rightarrow 0.125 \rightarrow 0.063 \rightarrow 0.000$). This marks the percolation transition boundary.

7.2 Tile-size sensitivity

Four tile heights at $k^2 = 26$, $R \in [1\text{M}, 1.03\text{M}]$, 32 strips:

Tile height does not significantly shift $f(r)$ estimates. The mean and extrema are consistent across two orders of magnitude in tile area. Larger tiles reduce variance (more primes per tile) but do not shift the mean.

Tile	Shells	$f_{\min}$	$f_{\max}$	Time/shell
200×200	150	0.281	0.750	0.7s
500×500	60	0.250	0.625	3.0s
1000×1000	30	0.313	0.594	11.1s
2000×2000	15	0.313	0.594	41.9s

Table 13: Tile-size sensitivity at $k^2 = 26$.

7.3 $k^2 = 40$ results

ISE probes at $k^2 = 40$ with $M = 64$ strips, 2000×2000 tiles, from $R = 600\text{M}$ to $R = 3.5\text{B}$:

R	$f(r)$	Regime
600M	1.000	Connected
700M	0.906	Declining
800M	0.609	Transition
900M	0.391	Transition
1.0B	0.203	Near collapse
1.2B	0.078	Sparse crossings
1.5B	0.016	Near extinction
2.0B	0.000	Extinct
2.5B	0.000	Extinct
3.0B	0.000	Extinct
3.5B	0.000	Extinct

Table 14: ISE $f(r)$ profile for $k^2 = 40$.

The $f(r)$ profile is a sigmoid with:

- $R_{0.5} \approx 839\text{M} \pm 12\text{M}$ (95% CI from logistic fit)
- First $f(r) = 0$ shells at $R \approx 2.0\text{B}$
- Total extinction from 2.0B onward

The logistic fit uses three calibration points in the transition region ($R = 700\text{M}, 800\text{M}, 900\text{M}$). Alternative functional forms (complementary error function, asymmetric sigmoid) have not yet been compared; model-selection sensitivity is an open question.

8 Open Questions

1. **Finite-size scaling.** Does $R_{0.5}$ depend on tile size W ? Data at $W \in \{500, 1000, 2000, 4000\}$ exists for $k^2 = 26$ but formal extrapolation to $W \rightarrow \infty$ has not been performed. The ISE moat estimate $R_{0.5}$ should be understood as a property of a specific tile geometry, not intrinsically of G_k . A finite-size scaling protocol—measuring $R_{0.5}(W)$ for several widths and extrapolating—would yield a tile-size-independent quantity.
2. **Percolation theory connection.** Is the tile connectivity transfer operator related to existing operators in percolation theory (e.g., the transfer matrix for site percolation on $\mathbb{Z}^2$)? The analogy is suggestive—both compress connectivity information through a strip—but Gaussian primes are deterministic, not random, so the connection is heuristic.

3. **Optimal tile geometry.** Square tiles ($W = H$) are confirmed necessary for reliable IO crossing detection. Is there an optimal W for a given k^2 and radius R ? The correlation length ξ of the crossing process sets a natural scale: $W \gg \xi$ is needed for reliable detection, but $W \gg \xi$ wastes computation.
4. **GPU-parallel UB.** Can the UB method be made GPU-parallel with composition on device? The individual tile builds are embarrassingly parallel (one CUDA thread block per tile), but the horizontal composition step requires sequential union-find merging across N_ℓ tiles. A hierarchical super-tile aggregation could reduce this to $O(\log N_\ell)$ sequential composition steps.
5. **Logistic model validation.** The sigmoid fit to $f(r)$ data is based on a logistic function fitted to a narrow transition window. The complementary error function $\frac{1}{2} \operatorname{erfc}(\cdot)$ is arguably more physically motivated for a percolation-like transition. Fitting both models and comparing $R_{0.5}$ estimates via AIC/BIC would quantify model-choice uncertainty.
6. **Tile-size dependence of $R_{0.5}$.** Narrower tiles measure lower $f(r)$ (they miss lateral paths), shifting the sigmoid leftward. If $R_{0.5}$ at $W = 2000$ and $R_{0.5}$ at $W = 4000$ differ by more than the CI, the estimation framework requires finite-size scaling corrections. This is the deepest methodological concern.

A Notation Summary

Symbol	Definition
$\mathbb{Z}[i]$	Gaussian integers
P	Set of all Gaussian primes
$G_k = (P, E_k)$	Gaussian prime graph at step bound k^2
Σ_k	Seed set: primes with $ \pi ^2 \leq k^2$
$C_0(k)$	Origin component of G_k
$R_{\text{moat}}(k)$	Moat radius: supremum of radii reached by $C_0(k)$
$T(a_0, b_0, W, H)$	Tile: lattice rectangle $[a_0, a_0+H] \times [b_0, b_0+W]$
$c = \lceil \sqrt{k^2} \rceil$	Collar width
T^+	Expanded tile (interior + collar of width c)
$\mathcal{I}, \mathcal{O}, \mathcal{L}, \mathcal{R}$	Inner, Outer, Left, Right faces
$\text{span}_T(X, Y)$	Count of components touching both faces X and Y
$\text{io_count}(T)$	$= \text{span}_T(\mathcal{I}, \mathcal{O})$
S_j	Strip j : vertical stack of tiles at lateral offset b_j
M	Number of strips
$f(r)$	Fraction of strips with $\text{io_count} > 0$ at radius r
$R_{0.5}$	Radius where $f(r) = 0.5$ (logistic midpoint)
$\rho(R)$	Local Gaussian prime density at radius R
$\mathcal{A}$	First-octant annular strip
$\mathcal{A}_n$	Radial stripe n within the annular strip
B_n	Band operator: horizontally composed tiles within stripe n
B	Full composed operator for the annular strip

Table 15: Notation summary.

B Algorithm Pseudocode (ISE)

B.1 Tile operator computation

Algorithm 1: ComputeTile(a_0, b_0, W, H, k^2)

Input: Tile bounds (a_0, b_0, W, H) , step bound k^2

Output: TileOperator (component face sets, face ports, io_count)

```

1  $c \leftarrow \lceil \sqrt{k^2} \rceil$ ;
2 Enumerate all Gaussian primes  $\pi$  in the expanded rectangle
    $[a_0 - c, a_0 + H + c] \times [b_0 - c, b_0 + W + c]$ ;
3 Initialize union-find UF on  $|\text{primes}|$  elements;
4 Build spatial hash: for each  $\pi$ , insert into cell grid with cell size  $c$ ;
5 foreach prime  $\pi$  at  $(a, b)$  do
6   foreach neighbor cell within distance 2 cells do
7     foreach prime  $\pi'$  in that cell with  $\text{index}(\pi') > \text{index}(\pi)$  do
8       if  $|\pi - \pi'|^2 \leq k^2$  then
9         UF.Union( $\text{index}(\pi)$ ,  $\text{index}(\pi')$ );
10 foreach prime  $\pi = (a, b)$  in the interior tile  $[a_0, a_0 + H] \times [b_0, b_0 + W]$  do
11   root  $\leftarrow$  UF.Find( $\text{index}(\pi)$ );
12   component  $\leftarrow$  ComponentID(root);
13   if  $a - a_0 \leq c$  then tag component with  $\mathcal{I}$ ; add face port;
14   if  $(a_0 + H) - a \leq c$  then tag component with  $\mathcal{O}$ ; add face port;
15   if  $b - b_0 \leq c$  then tag component with  $\mathcal{L}$ ; add face port;
16   if  $(b_0 + W) - b \leq c$  then tag component with  $\mathcal{R}$ ; add face port;
17 io_count  $\leftarrow |\{C : \mathcal{I} \in \text{faces}(C) \text{ and } \mathcal{O} \in \text{faces}(C)\}|$ ;
18 return TileOperator;

```

B.2 ISE ensemble evaluation

Algorithm 2: ISE($k^2, R_{\min}, R_{\max}, W, H, M$)

Input: Step bound k^2 , radial range $[R_{\min}, R_{\max}]$, tile dimensions $W \times H$, strip count M

Output: $f(r)$ profile for each shell

```

1  $c \leftarrow \lceil \sqrt{k^2} \rceil$ ;
2 stride  $\leftarrow W + 2c$ ;
3 offsets  $\leftarrow [c + j \cdot \text{stride} \text{ for } j = 0, \dots, M-1]$ ;
4 shells  $\leftarrow [(a_{\text{lo}}, a_{\text{hi}}) \text{ for } a_{\text{lo}} \in \{R_{\min}, R_{\min} + H, \dots, R_{\max} - H\}]$ ;
5 foreach shell  $(a_{\text{lo}}, a_{\text{hi}})$  do
6   io_counts  $\leftarrow [0] \times M$ ;
7   for  $j = 0$  to  $M - 1$  do // in parallel
8      $T \leftarrow \text{ComputeTile}(a_{\text{lo}}, \text{offsets}[j], W, a_{\text{hi}} - a_{\text{lo}}, k^2)$ ;
9     io_counts[j]  $\leftarrow \text{io\_count}(T)$ ;
10   $f(r) \leftarrow |\{j : \text{io\_counts}[j] > 0\}| / M$ ;
11  if  $f(r) = 0$  then flag shell as moat candidate;
12 return  $f(r)$  profile;

```

B.3 Compensated ISE

Algorithm 3: CompensatedISE($k^2, R_{\text{target}}, W, H, M, N_{\text{shells}}$)

Input: Step bound k^2 , target radius R_{target} , tile dimensions $W \times H$, strip count M , shells per probe N_{shells}

Output: $f(r)$ profile

```

1  $c \leftarrow \lceil \sqrt{k^2} \rceil$ ;
2  $\text{stride} \leftarrow W + 2c$ ;
3  $\text{offsets} \leftarrow [c + j \cdot \text{stride} \text{ for } j = 0, \dots, M-1]$ ;
4 for  $j = 0$  to  $M - 1$  do
5    $b_{\text{center}} \leftarrow \text{offsets}[j] + W/2$ ;
6    $a_{\text{base}}[j] \leftarrow \lfloor \sqrt{R_{\text{target}}^2 - b_{\text{center}}^2} \rfloor$ ;
7 for  $n = 0$  to  $N_{\text{shells}} - 1$  do
8    $\text{io\_counts} \leftarrow [0] \times M$ ;
9   for  $j = 0$  to  $M - 1$  do                                     // in parallel
10     $a_{\text{lo}} \leftarrow a_{\text{base}}[j] + n \cdot H$ ;
11     $T \leftarrow \text{ComputeTile}(a_{\text{lo}}, \text{offsets}[j], W, H, k^2)$ ;
12     $\text{io\_counts}[j] \leftarrow \text{io\_count}(T)$ ;
13   $f(r) \leftarrow |\{j : \text{io\_counts}[j] > 0\}|/M$ ;
14 return  $f(r)$  profile;
```

B.4 Fat-stripe campaign

Algorithm 4: FatStripe($k^2, r_{\min}, r_{\max}, W, H, C$)

Input: Step bound k^2 , annular radii $[r_{\min}, r_{\max}]$, tile dimensions $W \times H$, column-chunk size C

Output: blocked (boolean), spanning_count

```

1  $c \leftarrow \lceil \sqrt{k^2} \rceil$ ;
2 sieve_table  $\leftarrow$  Precompute( $L = 110,000$ );
3 composed  $\leftarrow$  null;
4 for  $a_{lo} = r_{\min}$  to  $r_{\max}$  step  $H$  do
5    $a_{hi} \leftarrow \min(a_{lo} + H, r_{\max})$ ;
6    $b_{\max} \leftarrow \min(\lceil \sqrt{r_{\max}^2 - a_{lo}^2} \rceil, a_{hi})$ ;
7   stripe_op  $\leftarrow$  null;
8   for  $b_{lo} = 0$  to  $b_{\max}$  step  $C \cdot W$  do
9      $b_{hi} \leftarrow \min(b_{lo} + C \cdot W, b_{\max})$ ;
10    // Phase 1: Sieve expanded chunk region (parallel over rows)
    primes  $\leftarrow$  SieveChunkRows( $a_{lo} - c, a_{hi} + c, b_{lo} - c, b_{hi} + c, \text{sieve\_table}$ );
    // Phase 2: Partition primes into virtual tiles with collar
    overlap
11    tile_primes  $\leftarrow$  Partition(primes,  $b_{lo}, W, c$ );
    // Phase 3: Sparse UF per tile  $\rightarrow$  TileOperator (parallel)
12    for each virtual tile  $t$  do // in parallel
13       $\lfloor$  tile_ops[ $t$ ]  $\leftarrow$  BuildTile( $a_{lo}, a_{hi}, b_t, b'_t, k^2, \text{tile\_primes}[t]$ );
    // Phase 4: Horizontal composition L $\rightarrow$ R within chunk
14    chunk_op  $\leftarrow$  tile_ops[0];
15    for  $t = 1$  to  $|\text{tile\_ops}| - 1$  do
16       $\lfloor$  chunk_op  $\leftarrow$  ComposeHorizontal(chunk_op, tile_ops[ $t$ ],  $k^2$ );
17    stripe_op  $\leftarrow$  ComposeHorizontal(stripe_op, chunk_op,  $k^2$ );
18  composed  $\leftarrow$  ComposeVertical(composed, stripe_op,  $k^2$ );
  // Verdict: radius-based spanning check
19 spanning_count  $\leftarrow$  0;
20 foreach component  $C$  in composed do
21   has_inner  $\leftarrow \exists p \in C : \sqrt{p.a^2 + p.b^2} \leq r_{\min} + c$ ;
22   has_outer  $\leftarrow \exists p \in C : \sqrt{p.a^2 + p.b^2} \geq r_{\max} - c$ ;
23   if has_inner  $\wedge$  has_outer then spanning_count  $+= 1$ ;
24 blocked  $\leftarrow$  (spanning_count = 0);
25 return blocked, spanning_count;

```

C Feasibility Calculation Details

C.1 Neighbor counts by k^2

C.2 Percolation threshold estimation for $k^2 = 40$

At $k^2 = 36$, the percolation boundary ($f(r) = 0$) was found at $R \approx 80.4\text{M}$ with empirical prime density $\rho \approx 3.50\%$. The mean degree at the transition:

$$d_c = \rho \times z = 0.0350 \times 112 = 3.92.$$

k^2	Nonzero neighbor vectors	Collar c	Max-excursion vector
2	4	2	(1, 1)
26	80	6	(5, 1)
32	92	6	(4, 4)
36	112	6	(6, 0)
40	128	7	(6, 2)

Table 16: Neighbor counts and collar widths by step bound.

Assuming constant critical mean degree across k^2 values:

$$\rho_c(k^2 = 40) = \frac{d_c}{z} = \frac{3.92}{128} = 0.0306 = 3.06\%.$$

Setting $\rho(R) = C/(2 \ln R) = \rho_c$:

$$\ln R = \frac{C}{2\rho_c} = \frac{1.274}{2 \times 0.0306} = 20.82, \quad R = e^{20.82} \approx 1.10 \times 10^9.$$

This central estimate of $R \approx 1.1\text{B}$ is consistent with the empirical sigmoid midpoint $R_{0.5} \approx 839\text{M}$ (the ISE measures local percolation, which transitions earlier than the mean-degree prediction).

C.3 Memory requirements

Per-tile working memory for 2000×2000 tile with $c = 7$:

Component	Size
Expanded tile bitmap (2015^2)	507 KB
Union-find parent array (n u32)	~ 16 MB
Union-find rank array (n u8)	~ 4 MB
Spatial hash	~ 2 MB
Total per tile	~ 23 MB

Table 17: Per-tile memory breakdown.

With 6 parallel tiles (Jetson Orin Nano, 6 cores): ~ 138 MB peak. With 64 parallel tiles (cloud): ~ 1.5 GB.

C.4 Tsuchimura comparison

A Tsuchimura-style full sweep to $R = 1.5\text{B}$ at $k^2 = 40$ would require storing all primes in the disk of radius 1.5B :

$$N_{\text{primes}} \approx \frac{\pi R^2 \cdot \rho(R)}{1} = \frac{\pi \cdot (1.5 \times 10^9)^2 \cdot 0.031}{1} \approx 2.2 \times 10^{17}.$$

At 16 bytes per prime (coordinates + component ID): ~ 3.5 PB. Union-find adds comparable storage. This is infeasible on any existing hardware.

The ISE, by contrast, processes one 2000×2000 tile at a time with ~ 23 MB working memory, achieving a memory reduction of approximately $10^{11} \times$.

C.5 Primality validity ceiling

The implementation uses a 9-witness deterministic Miller–Rabin test, valid for all $n < 3.825 \times 10^{18}$. At radius R , the maximum norm tested is approximately R^2 . The hard ceiling is $R \leq 1.955 \times 10^9$ ($\approx 1.955\text{B}$). Beyond this, either additional witnesses or a BPSW test is required. The $k^2 = 40$ campaign data through $R = 3.5\text{B}$ uses 12 witnesses for probes beyond the 9-witness ceiling.

C.6 GPU throughput calculations

The throughput projections in Section 5.6 are derived as follows.

Sequential per-tile time. On Jetson Orin Nano, the GPU primality kernel takes ~ 170 ms for a 2000×2000 tile. Eliminating the CPU UF stage and D2H copy removes ~ 42 ms, but the GPU UF kernel adds ~ 50 ms (estimate based on ECL-CC benchmarks on similar grid sizes), yielding ~ 180 ms total per tile.

Cross-GPU scaling. Scaling from Jetson to discrete GPUs uses the SM count and clock ratio:

$$t_{\text{GPU}} = t_{\text{Jetson}} \times \frac{\text{SM}_{\text{Jetson}}}{\text{SM}_{\text{GPU}}} \times \frac{f_{\text{Jetson}}}{f_{\text{GPU}}} \times \alpha,$$

where $\alpha \approx 0.7$ accounts for memory bandwidth differences (Jetson uses shared LPDDR5; discrete GPUs use dedicated HBM/GDDR6X with higher bandwidth).

Batched throughput. With tile-per-SM batching, the effective per-tile time is the sequential time divided by the SM count, assuming sufficient tiles to fill all SMs:

$$t_{\text{batched}} = \frac{t_{\text{sequential}}}{\text{SM count}}.$$

At $R = 1.5\text{B}$, the shell contains $\sim 589\text{K}$ tiles—far exceeding any GPU’s SM count—so the batching assumption holds.

Cost model. Cloud cost estimates use: RTX 4090 at $\sim \$2/\text{hr}$ (consumer cloud), A100 SXM4 at $\sim \$2/\text{hr}$ (reserved spot pricing). Multi-GPU configurations assume linear scaling (tile-level parallelism with no inter-GPU communication).

References

- [1] S. Jaiganesh and M. Burtscher, *A high-performance connected components implementation for GPUs*, in Proc. PPOPP, 2018, pp. 92–104.
- [2] M. Sutton, T. Ben-Nun, and A. Barak, *Optimizing parallel graph connectivity computation via subgraph sampling*, in Proc. SC, 2018, Article 14.